

Vision — Features: Now & Next

What farmers can do today across the website and mobile app, and a forward roadmap of more advanced capabilities we could build next.

Shipped today

Near-term upgrades

Advanced future

Web + Android APK live

1. What Vision is today

Vision is a complete farmer loop: create an account → manage camps → run an explainable rangeland assessment → ask follow-ups → compare camps — with live weather, research plot matching, optional photos, and offline browse of saved data.

WEBSITE

<https://vision-kf45.vercel.app>

API

<https://vision-52yf.onrender.com>

MOBILE

Expo EAS Android APK

Demo login: **demo@vision.na** / **vision123** — Otjiwarongo Demonstration Farm with North, River, Acacia, and Rest camps.

2. Features shipped now

NOW Auth & accounts

- Register + login (JWT)
- Auto-creates a farm on signup
- Optional farm name & region
- Secure token storage on mobile
- Seeded demo farmer account

NOW Farm & camps

- Create / edit camps (area, livestock, grazing)
- Coords: map picker (web) / GPS (mobile)
- Camp history of past assessments
- Onboarding when no camps yet
- Offline queue for create/update camp

NOW Assessments

- Guided wizard: Camp → Herd → Grazing → Photos → Review
- Statuses: Good / Watch / High concern / Insufficient data
- Evidence, confidence, limitations, next steps
- Gemini tool-calling agent
- Rule-based fallback if AI is down

NOW AI advisor (Ask)

- Conversational grazing advice
- Uses same farm/weather/research tools
- Suggested starter questions
- Text-to-speech (web + mobile)
- Speech-to-text on website

NOW Data & weather

- Live Open-Meteo rainfall & temperature
- ~70 Namibia research plots matched
- LSU / ha & grazing-duration calcs
- Comparable plot cards + cover trends
- Drought-context hints

NOW Photos

- Optional N/E/S/W camp photos
- Camera + library on mobile
- Cautious vision analysis (density, soil, dryness...)
- Never invents exact biomass %
- Findings strengthen the assessment

NOW Compare

- Side-by-side camp comparison
- AI-assisted comparison narrative
- Useful for rotation planning

NOW Offline & PWA

- Cache farms, camps, past assessments
- Queue camp edits when offline
- Connectivity banner (slow / offline)
- Website PWA shell installable
- Assess / Ask / weather need network

NOW Deploy surfaces

- Website on Vercel
- API on Render (Docker)
- DB on Supabase
- Android APK via Expo EAS
- Shared design system (sand / veld)

3. Website vs mobile parity (today)

Capability	Website	Mobile (APK / Expo)
Auth, camps, assess, ask, compare	YES	YES
Offline cache + camp edit queue	YES (+ PWA)	YES
Set camp location	Map picker	GPS "Use my location"
Voice input (STT)	YES (browser)	Limited (not Expo Go native)
Voice playback (TTS)	YES	YES
Photos	File upload	Camera + library

4. Near-term upgrades (practical next steps)

High-value improvements that fit the current architecture without rewriting the core.

NEXT Product polish

- Password reset + email verification
- Delete camp / archive camp in the UI
- Multi-farm switcher for larger operators
- Native speech-to-text on Android APK
- iOS build via EAS (TestFlight / App Store)
- Push notifications when an assessment finishes (cold API wake)
- Export assessment as PDF / WhatsApp share card
- Afrikaans / Oshiwambo / Otjiherero language packs

NEXT Reliability & field use

- Offline assessment draft (queue full assess when back online)
- Prefetch weather when online to soften weak-signal areas
- SMS / USSD light mode for feature phones
- Low-bandwidth “text-only” assessment summary
- Stronger cold-start UX while Render wakes
- Play Store listing + signed production release
- Admin seed refresh button for new research plots

5. Advanced features (future roadmap)

Larger capabilities that deepen the science, collaboration, and decision support — building on the same agent + evidence model.

ADVANCED Sensing & models

- Satellite NDVI / fractional cover layers per camp
- Seasonal carrying-capacity models by ecoregion
- Rainfall forecast ensembles + drought early warning
- Herd growth / offtake planning scenarios
- Time-series: “how has this camp changed this season?”
- Drone / higher-res photo pipelines

ADVANCED Collaboration

- Multi-user farms (owner / herder / advisor roles)
- Extension-officer dashboard & review queue
- Shared cooperatives / communal grazing units
- Audit trail of advice accepted vs overridden
- Referral workflows to MAWLR / local officers
- Anonymised regional benchmarking

ADVANCED Intelligence

- Proactive weekly “camp health” briefs
- Rotation planner that suggests move dates
- Multimodal chat with live map overlays
- Fine-tuned local language / dialect models
- Human-in-the-loop validation of photo labels
- Confidence calibration from field outcomes

ADVANCED Integrations

- Livestock market price feeds (offtake timing)
- Government / NGO reporting exports
- IoT trough / fence sensors (optional)
- Open data APIs for researchers (opt-in)
- WhatsApp Business bot for Ask + alerts

ADVANCED Trust & scale

- Field validation studies with farmers & officers
- Explainability report per assessment (tool trace)
- Region-specific rule packs (Kalahari, Highland...)
- Privacy controls & data residency options
- Multi-country expansion with local seed datasets

6. Suggested priority order

Phase	Focus	Examples
Now	Hackathon-ready product	Auth, camps, assess, ask, compare, weather, research match, photos, offline browse, web + Android APK
Next	Field readiness	iOS, native STT, PDF/WhatsApp share, password reset, delete camp, offline assess queue, languages
Later	Advanced impact	Satellite layers, multi-user + extension dashboard, rotation planner, SMS/WhatsApp channel, validation studies

Vision’s design already separates *farmer data*, *live weather*, *research plots*, *calculations*, and *AI conclusions* — so future sensors and models can plug in as new evidence sources without changing the honesty model.